



MATLAB Drive > dsp4_1.m

```
1 clc;
2 clear;
3 close all;
4
5 %% --- Input Parameters ---
6 fp = 1000;           % Passband cutoff (Hz)
7 fs = 4000;           % Sampling frequency (Hz)
8 N = 50;              % Filter order (even number pr
9
10 wc = fp/(fs/2); % Normalized cutoff frequency
11
12 %% --- FIR Low Pass using Rectangular Window ---
13 b_rect = fir1(N, wc, 'low', rectwin(N+1));
14
15 disp('--- Rectangular Window Coefficients ---')
16 disp(b_rect);
17
18 %% --- FIR Low Pass using Hamming Window ---
19 b_hamm = fir1(N, wc, 'low', hamming(N+1));
20
21 disp('--- Hamming Window Coefficients ---');
22 disp(b_hamm);
23
24 %% --- FIR Low Pass using Hanning Window ---
25 b_hann = fir1(N, wc, 'low', hanning(N+1));
26
27 disp('--- Hanning Window Coefficients ---');
28 disp(b_hann);
29
30 %% --- Plot Frequency Responses ---
31 figure;
32 freqz(b_rect,1);
33 title('Low-pass FIR using Rectangular Window')
34
35 figure;
36 freqz(b_hamm,1);
37 title('Low-pass FIR using Hamming Window');
38
39 figure;
40 freqz(b_hann,1);
41 title('Low-pass FIR using Hanning Window');
```

Figure 1: dsp4_1

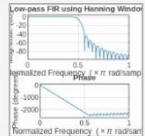
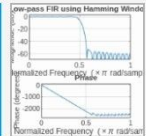
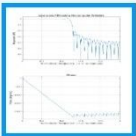
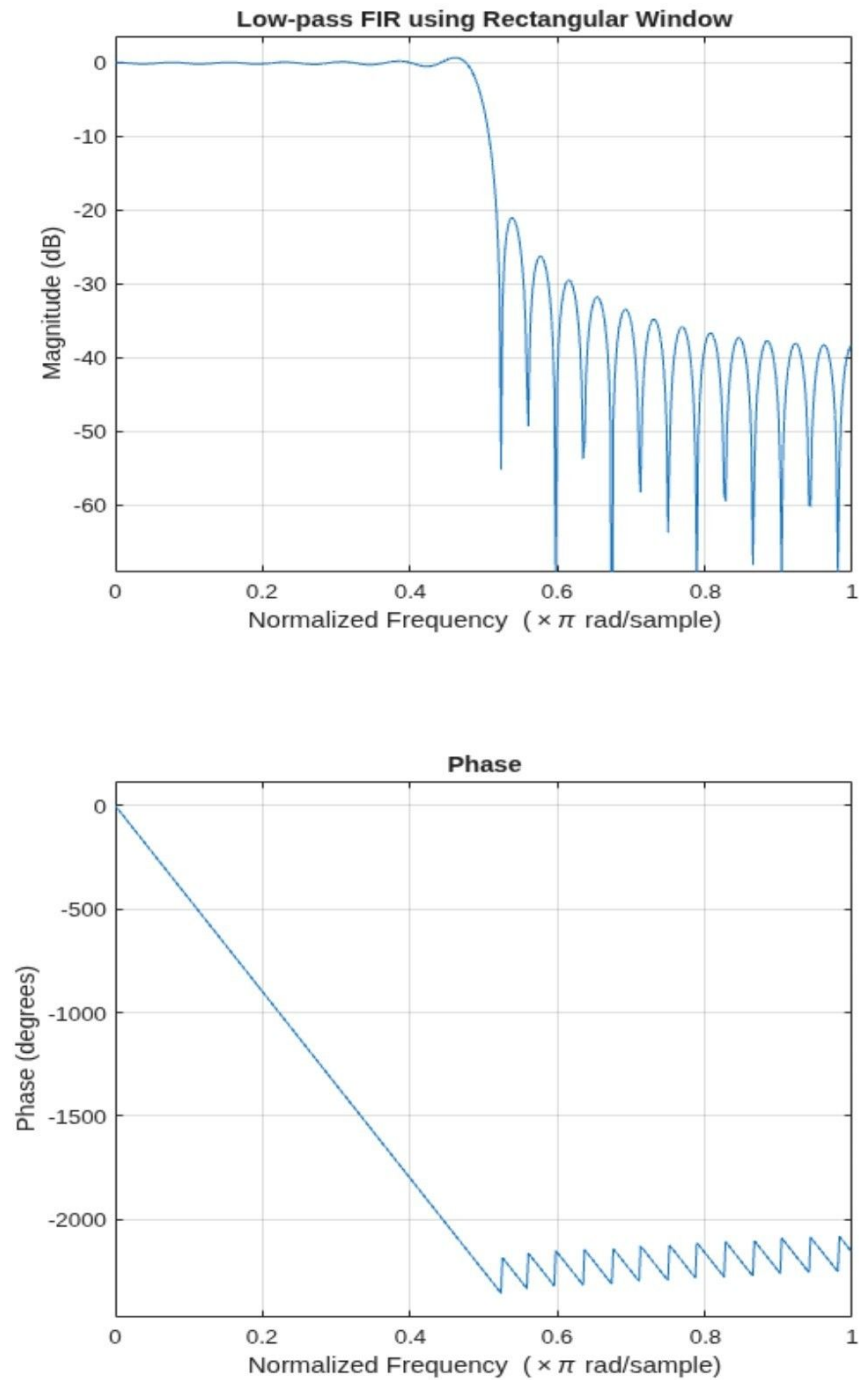


Figure 2: dsp4_1

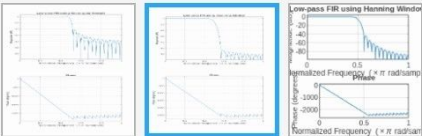
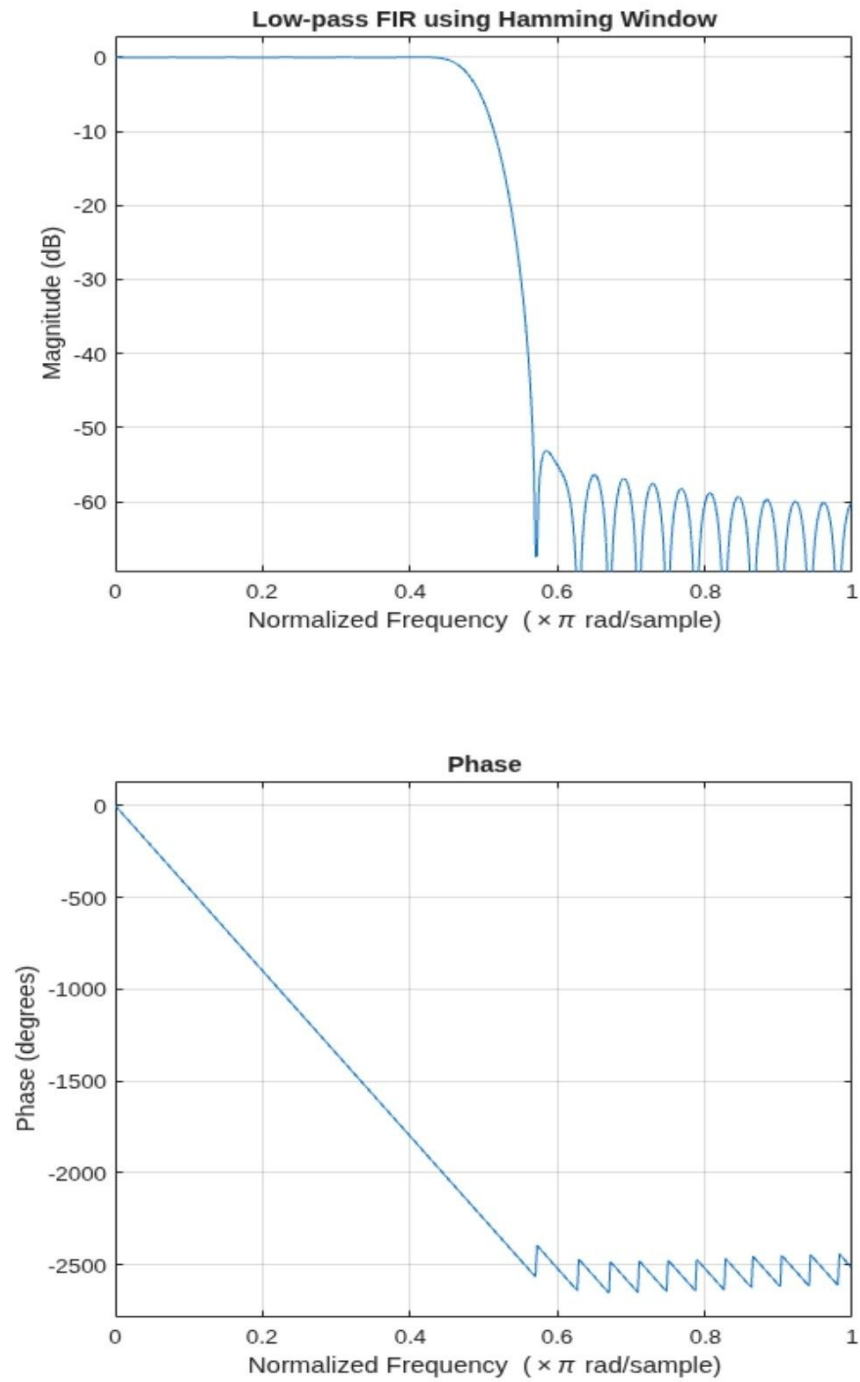
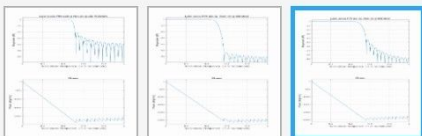
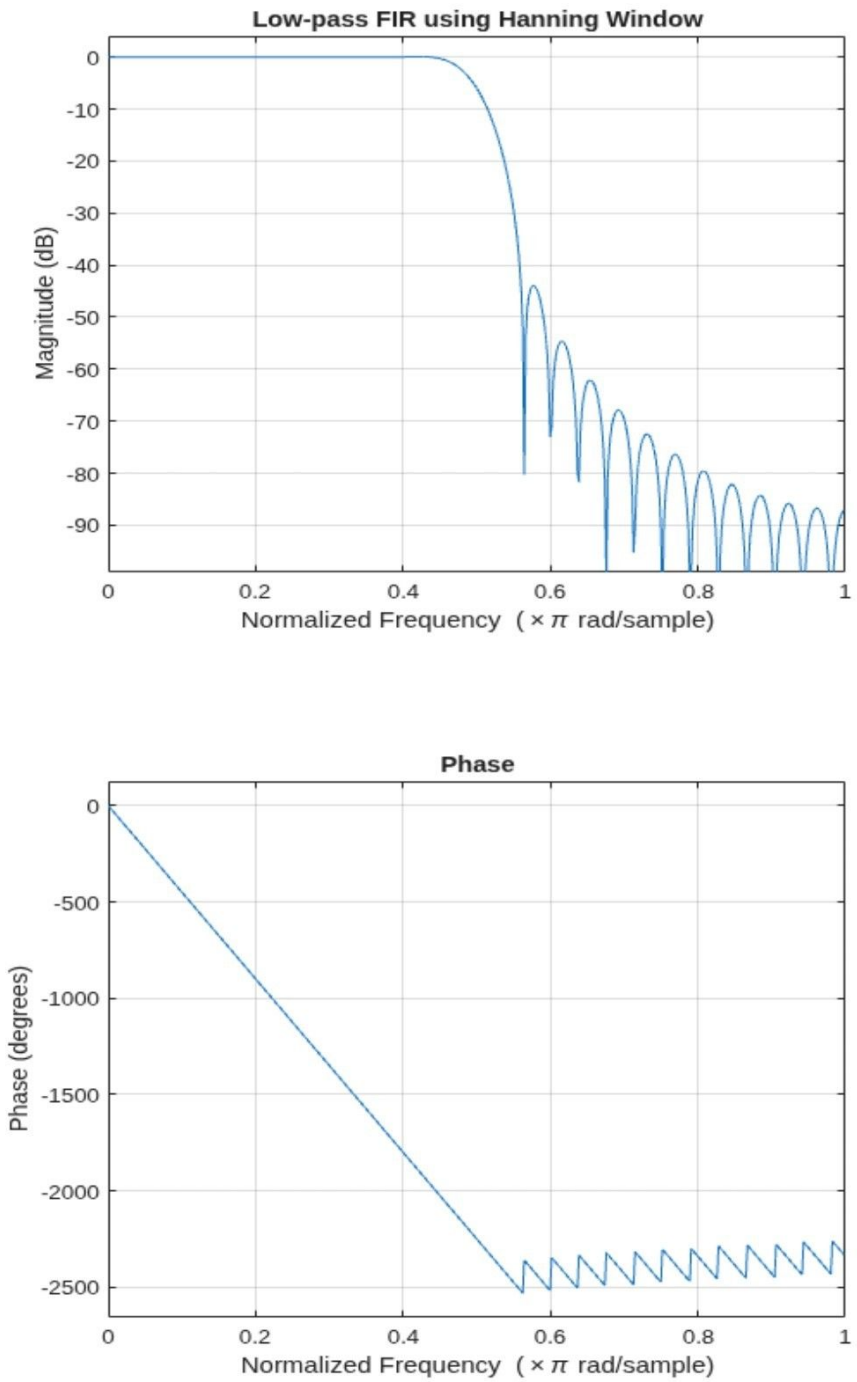


Figure 3: dsp4_1





MATLAB Drive > dsp4_2.m

```
1 clc;
2 clear;
3 close all;
4
5 %% --- Input Parameters ---
6 fp = 1000;          % Passband cutoff (Hz)
7 fs = 4000;          % Sampling frequency (Hz)
8 N = 50;             % Filter order (even number pr
9
10 wc = fp/(fs/2); % Normalized cutoff frequency
11
12 %% --- FIR Low Pass using Hamming Window ---
13 b_hamm = fir1(N, wc, 'low', hamming(N+1));
14
15 disp('--- Hamming Window Coefficients ---');
16 disp(b_hamm);
17
18 %% --- Plot Frequency Response ---
19 figure;
20 freqz(b_hamm,1);
21 title('Low-pass FIR using Hamming Window');
```

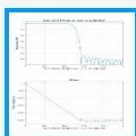
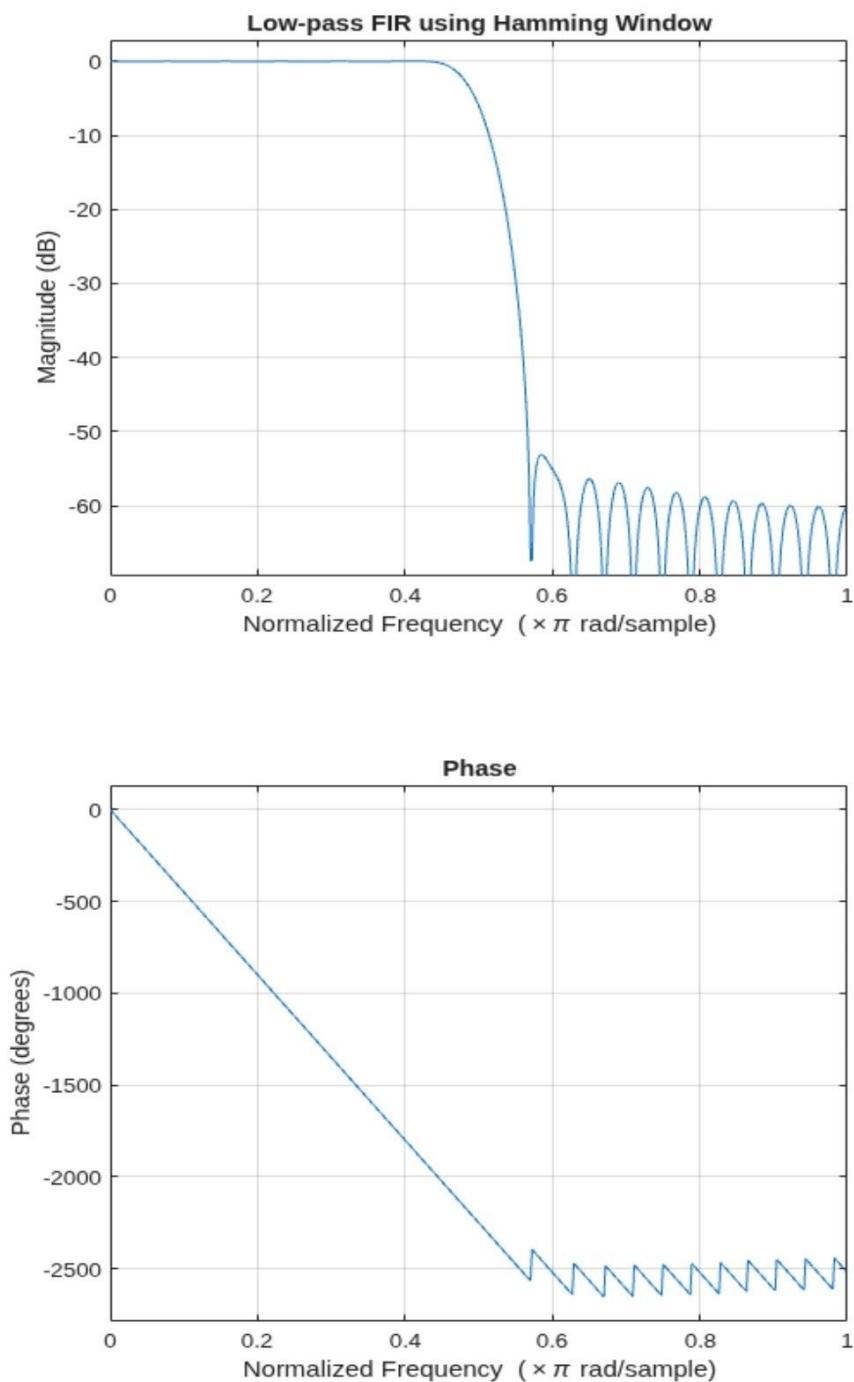


Figures

SHOW DATA CURSOR



Figure 1: dsp4_2





MATLAB Drive > dsp4_3.m

```
1 clc;
2 clear;
3 close all;
4
5 %% --- Input Parameters ---
6 fp = 1000;          % Passband cutoff (Hz)
7 fs = 4000;          % Sampling frequency (Hz)
8 N = 50;             % Filter order (even number pr
9
10 wc = fp/(fs/2); % Normalized cutoff frequency
11
12 %% --- FIR Low Pass using Hanning Window ---
13 b_hann = fir1(N, wc, 'low', hanning(N+1));
14
15 disp('--- Hanning Window Coefficients ---');
16 disp(b_hann);
17
18 %% --- Plot Frequency Response ---
19 figure;
20 freqz(b_hann,1);
21 title('Low-pass FIR using Hanning Window');
```



Figures

SHOW DATA CURSOR



Figure 1: dsp4_3

